

三代测序分析-nanopore

1.软件环境

```
# packages in environment at /home/jj2024/anaconda3/envs/nanopore:
#
# Name                                Version                                Build                                Channel
_libgcc_mutex                         0.1                                    conda_forge                         conda-forge
_openmp_mutex                         4.5                                    2_gnu                              conda-forge
bwa                                   0.7.18                                he4a0461_1                          bioconda
bzip2                                 1.0.8                                h4bc722e_7                          conda-forge
c-ares                               1.33.0                                ha66036c_0                          conda-forge
ca-certificates                      2024.7.4                              hbcca054_0                          conda-forge
dos2unix                             7.5.2                                ha770c72_3                          conda-forge
gettext                              0.22.5                                he02047a_3                          conda-forge
gettext-tools                        0.22.5                                he02047a_3                          conda-forge
homer                               4.11                                  pl5262h4ac6f70_9                    bioconda
htslib                               1.20                                  h5efdd21_2                          bioconda
keyutils                             1.6.1                                h166bdaf_0                          conda-forge
krb5                                 1.21.3                               h659f571_0                          conda-forge
ld_impl_linux-64                    2.40                                  hf3520f5_7                          conda-forge
libasprintf                          0.22.5                                he8f35ee_3                          conda-forge
libasprintf-devel                   0.22.5                                he8f35ee_3                          conda-forge
libcurl                              8.8.0                                  hca28451_1                          conda-forge
libdeflate                           1.21                                  h4bc722e_0                          conda-forge
libedit                              3.1.20191231                         he28a2e2_2                          conda-forge
libev                                 4.33                                  hd590300_2                          conda-forge
libffi                               3.4.2                                h7f98852_5                          conda-forge
libgcc-ng                           14.1.0                                h77fa898_0                          conda-forge
libgettextpo                        0.22.5                                he02047a_3                          conda-forge
libgettextpo-devel                  0.22.5                                he02047a_3                          conda-forge
libgomp                              14.1.0                                h77fa898_0                          conda-forge
libidn2                              2.3.7                                  hd590300_0                          conda-forge
libnghttp2                           1.58.0                                h47da74e_1                          conda-forge
libnsl                               2.0.1                                  hd590300_0                          conda-forge
libsqlite                             3.46.0                                hde9e2c9_0                          conda-forge
libssh2                              1.11.0                                h0841786_0                          conda-forge
libstdc++-ng                         14.1.0                                hc0a3c3a_0                          conda-forge
libunistring                         0.9.10                                h7f98852_0                          conda-forge
libuuid                              2.38.1                                h0b41bf4_0                          conda-forge
libxcrypt                            4.4.36                                hd590300_1                          conda-forge
libzlib                              1.2.13                                h4ab18f5_6                          conda-forge
ncurses                              6.5                                    h59595ed_0                          conda-forge
openssl                              3.3.1                                  hb9d3cd8_3                          conda-forge
perl                                 5.32.1                                7_hd590300_perl5                    conda-forge
pip                                  24.2                                  pyhd8ed1ab_0                        conda-forge
python                              3.11.0                                he550d4f_1_cpython                  conda-forge
readline                             8.2                                    h8228510_1                          conda-forge
samtools                             1.20                                  h50ea8bc_1                          bioconda
seqkit                               2.8.2                                  h9ee0642_1                          bioconda
setuptools                           72.1.0                                pyhd8ed1ab_0                        conda-forge
subread                              2.0.6                                  he4a0461_2                          bioconda
tk                                   8.6.13                                noxft_h4845f30_101                  conda-forge
tzdata                              2024a                                  h0c530f3_0                          conda-forge
unzip                                6.0                                    h7f98852_3                          conda-forge
```

wget	1.21.4	hda4d442_0	conda-forge
wheel	0.44.0	pyhd8ed1ab_0	conda-forge
xz	5.2.6	h166bdaf_0	conda-forge
zlib	1.2.13	h4ab18f5_6	conda-forge
zstd	1.5.6	ha6fb4c9_0	conda-forge

2.对三代下机数据进行pod5文件或者fast5类型转换为fastq

```
##guppy碱基识别
./guppy/bin/guppy_basecaller -i /dir/fast5/ -s /dir/fastq/ -r --min_qscore 0 --
flowcell FLO-MIN106 --kit SQK-RNA002 --num_callers 1 --cpu_threads_per_caller 8
##dorado碱基识别
dorado basecaller /home/jj2024/.dorado/models/rna004_130bps_hac@v5.1.0 pod5/ --
emit-fastq > 20241107_sample5.fastq
```

3.RNADNA转换

```
seqkit seq ./rna.fq --rna2dna > dna.fastq
```

4.碱基质量筛选（去除q<6的）

```
seqkit seq -Q 6 -m 80 ${n}.fq | gzip -c >${n}.fq.gz
```

5.比对tRNA序列（正常到这一步，linux上的就结束了）

```
#!/bin/sh
ref="/home/jj2024/00.db/01.hg38/08.hg38_trna_nomt_3adapter/human_tRNA_nomt_3adapt
er.fa"

for i in *.fq;do
    n=${i/.fq/}
    bwa mem -w13 -k6 -xont2d -T20 $ref ${n}.fq |\
    samtools view -q1 -F4 -b - |\
    samtools sort - > ${n}.bam
    samtools index ${n}.bam
    samtools idxstats ${n}.bam >${n}_trna_stat_q1.txt
done
```

####接下来的步骤是用于探寻自建库数据，未比对上tRNA参考基因组的序列是否比对到了全基因组或者adapter上面

1.提取未必对序列数据

```
#!/bin/sh
ref="/home/jj2024/00.db/01.hg38/02.hg38_trna_mt/human_tRNA_mt.fa"

for i in *.fastq;do
    n=${i/.fastq/}
    bwa mem -w13 -k6 -xont2d -T20 $ref ${n}.fastq |\
    samtools view -f4 -b - |\
    samtools sort - > ${n}_mt.bam
```

```

        samtools index ${n}_mt.bam
done

for i in *bam;do
    n=${i/.bam/}
    samtools fastq -n ${n}.bam >${n}_unmapped.fq
done

```

2.比对全基因组

```

#!/bin/sh

ref="/home/jj2024/00.db/01.hg38/06.hg38_dna_bwaidx/hg38_dna.fa"

for i in *.fq;do
    n=${i/.fq/}
    bwa mem -w13 -k6 -xont2d -T20 $ref ${n}.fq |\
    samtools view -F4 -b - |\
    samtools sort - > ${n}.bam
    samtools index ${n}.bam
done

bedtools genomecov -ibam ${n}.bam -bg > coverage.bedgraph

```

3.比对adapter

```

#!/bin/sh
ref="/home/jj2024/00.db/01.hg38/07.adapter_3/adapted_3.fa"

for i in *.fq;do
    n=${i/.fq/}
    bwa mem -w13 -k6 -xont2d -T20 -M $ref ${n}.fq |\
    samtools view -q0 -F2308 -b - |\
    samtools sort - > ${n}_adapter.bam
    samtools index ${n}_adapter.bam
    samtools idxstats ${n}_adapter.bam >${n}_adapter.txt
done

```

4.查看比对到adapter上的序列比对上了多少tRNA

```

#!/bin/sh

for i in *_adapter.bam;do
    n=${i/_adapter.bam/}
    samtools view -h ${n}_adapter.bam adapter1_adapter2_1|samtools view -bs -
|samtools fastq - >${n}_adapter12.fq
    samtools view -h ${n}_adapter.bam adapter1_1|samtools view -bs -
|samtools fastq - >${n}_adapter1.fq
    samtools view -h ${n}_adapter.bam adapter2_1|samtools view -bs -
|samtools fastq - >${n}_adapter2.fq
done

```

```
#!/bin/sh
ref="/home/jj2024/00.db/01.hg38/03.hg38_trna_nomt/human_tRNA_nomt.fa"

for i in *.fq;do
    n=${i/.fq/}
    bwa mem -w13 -k6 -xont2d -T20 -M $ref ${n}.fq |\
    samtools view -q0 -F2308 -b - |\
    samtools sort - > ${n}_nomt.bam
    samtools index ${n}_nomt.bam
    samtools idxstats ${n}_nomt.bam >${n}_trna_q0_nomt.txt
done
```